

Macros & VBA: Automate Your Work

Stop doing the same thing twice. Learn how to make Excel work for you — automatically.

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What's a Macro?

Automated Sequence

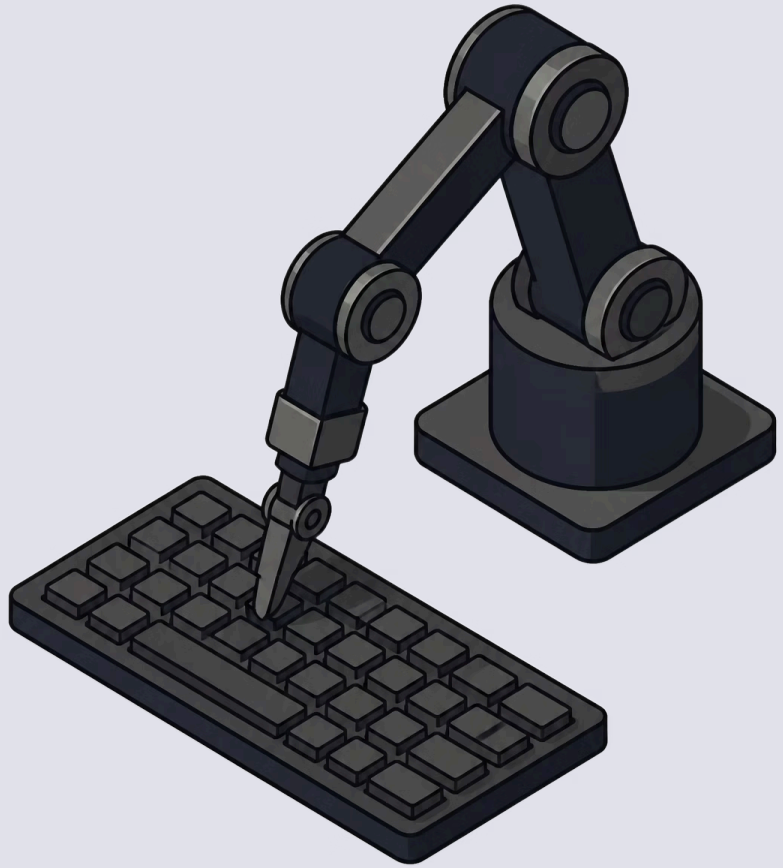
A recorded or written sequence of actions that Excel repeats automatically — no clicks required.

Eliminates Repetition

Automate formatting, calculations, and data entry that would otherwise take hours of manual effort.

Real Example

Format **500 rows of data in seconds** instead of hours — the same result, every single time.



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Why Learn Macros?



Speed

Eliminate manual, repetitive work. What takes minutes by hand runs in seconds with a macro.



Accuracy

The same steps execute identically every time — no typos, no missed rows, no human error.



Career Skill

Employers across finance, consulting, and operations actively seek candidates who can automate workflows.

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Recording vs. Writing Macros

Record a Macro

Click **Record Macro**, perform your actions in Excel, then stop recording. Excel writes the code automatically.

- Fast to create — no coding needed
- Great for simple, one-time tasks
- Limited: can't handle logic or variation

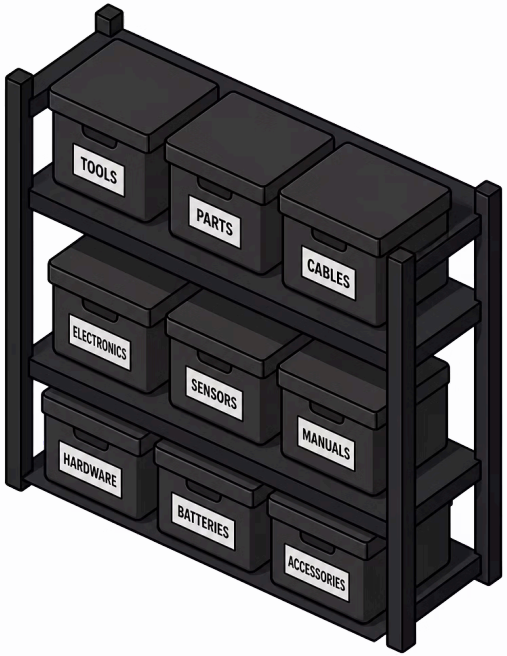
Write VBA Code

Open the VBA editor and write the logic yourself. More effort up front — far greater power in return.

- Handle conditions, loops, and user input
- Reusable across workbooks and teams
- The foundation for serious automation

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VBA Basics: Variables & Data Types



What Is a Variable?

Think of a variable as a **labeled box** — it holds a piece of data your code can use and update at any time.

Common Data Types

- **Integer** — whole numbers (e.g., row counts)
- **String** — text values (e.g., a name or label)
- **Boolean** — true or false flags

Declaring a Variable

```
Dim count As Integer
```

This creates a variable named `count` that stores whole numbers.

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VBA Basics: Loops & Conditions

Loops

Repeat a block of code multiple times without rewriting it.

```
For i = 1 To 10
  ' runs 10 times
Next i
```

Conditions

Run code only when a specific rule is true.

```
If x > 5 Then
  ' do something
End If
```

Together in Action

Loop through every row in a dataset, check each value with a condition, and take action only where it matches — automatically.

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Practical Example: Auto-Format a Report

2s

With a Macro

3 clicks. Bold headers, blue fill, resized columns — done instantly.

100

Minutes Saved

5 min per report × 20 reports = **100 minutes back** every cycle.

The Task

Every week you receive a raw data export. It needs bold headers, a blue header row, and auto-fitted column widths before sharing with your team.

The Manual Way

Select headers → bold → fill color → select all columns → auto-fit. Repeat for every single report.

The Macro Way

Record the steps once. Run the macro forever. **Same result, zero repetition.**

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Next Steps

01

Record Your First Macro

Use **Record Macro** in Excel to capture a simple task — watch how Excel translates your clicks into code.

02

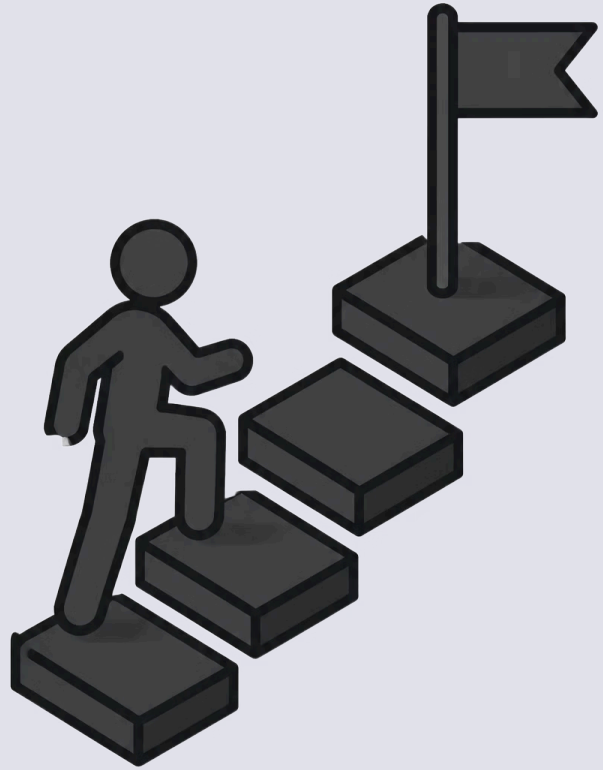
Read the Generated Code

Press **Alt+F11** to open the VBA editor. Study what Excel wrote — this is your first VBA lesson for free.

03

Write Your First Loop

Practice by writing a simple `For` loop that changes cell background colors based on their values.



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